MAT 5210 Homework 2 — Solutions

Problem 1

Let $L = \mathbb{Q}(\sqrt[3]{2})$ and $K = \mathbb{Q}$. Show $\mathrm{Aut}_K(L) = \{\mathrm{id}\}$.

Proof. Let $\alpha = \sqrt[3]{2} \in \mathbb{R}$ be the real cube root of 2, so $L = \mathbb{Q}(\alpha) \subseteq \mathbb{R}$. Let $\sigma \in \mathrm{Aut}_K(L)$. Since σ fixes $\mathbb{Q}$ and $\alpha^3 - 2 = 0$, we get $\sigma(\alpha)^3 = 2$, so $\sigma(\alpha)$ is a root of $x^3 - 2$ lying in L. The roots of $x^3 - 2$ in $\mathbb{C}$ are $\alpha,\ \alpha\zeta,\ \alpha\zeta^2$ with $\zeta = e^{2\pi i/3}$, and the latter two are non-real. Since $L \subseteq \mathbb{R}$, we must have $\sigma(\alpha) = \alpha$. As $L = \mathbb{Q}(\alpha)$ is generated by α over $\mathbb{Q}$, $\sigma = \mathrm{id}$.

Consequently $L^{\mathrm{Aut}_K(L)} = L^{\{\mathrm{id}\}} = L$. Since $x^3 - 2$ is irreducible over $\mathbb{Q}$ (Eisenstein at 2), $[L : \mathbb{Q}] = 3$, so $L \neq K$. Hence $L^{\mathrm{Aut}_K(L)} \neq K$ and $L : K$ is not Galois. ∎

Problem 2

Let $M : K$ be algebraic and $G \leq \mathrm{Aut}_K(M)$. Show $M : M^G$ is Galois.

Proof. Put $F = M^G = \{x \in M : \ g(x) = x \ \forall g \in G\}$. Recall that $M : F$ is Galois means $M^{\mathrm{Aut}_F(M)} = F$.

Every $g \in G$ fixes F pointwise by definition of F, so $G \leq \mathrm{Aut}_F(M)$. Taking fixed fields reverses inclusions:

$$M^{\mathrm{Aut}_F(M)} \subseteq M^G = F.$$

Conversely, every element of $\mathrm{Aut}_F(M)$ fixes F pointwise, so $F \subseteq M^{\mathrm{Aut}_F(M)}$. Therefore $M^{\mathrm{Aut}_F(M)} = F = M^G$, i.e. $M : M^G$ is Galois. ∎

Problem 3

Show that if $f \in K[x]$ has degree n, its splitting field has degree $\leq n!$ over K.

Proof. We prove by induction on $n \geq 0$ that for every field K and every $f \in K[x]$ of degree n, a splitting field S of f over K satisfies $[S : K] \leq n!$.

For $n = 0, 1$ the polynomial splits over K, so $S = K$ and $[S : K] = 1 \leq n!$.

Let $n \geq 2$ and assume the claim for degree $n - 1$ (over all fields). Let S be a splitting field of f over K and let $\alpha \in S$ be a root of f (if f has no root in S it would not split, so f has a root since $n \geq 1$). Then $[K(\alpha) : K] \leq n$, because the minimal polynomial of α divides f. Over $K(\alpha)$ we can write $f = (x - \alpha)f_1$ with $\deg f_1 = n - 1$. The field S is generated over K by the roots of f, i.e. by α and the roots of f_1, so S is generated over $K(\alpha)$ by the roots of f_1; since f splits in S, so does f_1. Thus S is a splitting field of f_1 over $K(\alpha)$, and by the induction hypothesis $[S : K(\alpha)] \leq (n - 1)!$. Hence

$$[S : K] = [S : K(\alpha)]\,[K(\alpha) : K] \leq (n - 1)! \cdot n = n!.$$ ∎

Problem 4

Degrees of splitting fields.

(a) $x^3 - 1 = (x-1)(x^2+x+1)$ over $\mathbb{Q}$. The roots are $1, \omega, \omega^2$ with $\omega = e^{2\pi i/3} = \frac{-1+\sqrt{-3}}{2}$. The splitting field is $\mathbb{Q}(\omega) = \mathbb{Q}(\sqrt{-3})$. The polynomial x^2+x+1 has no rational roots (its discriminant is $-3 < 0$), so it is irreducible over $\mathbb{Q}$ and

$$[\mathbb{Q}(\omega) : \mathbb{Q}] = \mathbf{2}.$$

(b) $x^3 - 2$ over $\mathbb{Q}$. With $\alpha = \sqrt[3]{2} \in \mathbb{R}$ and ω as above, the roots are $\alpha, \alpha\omega, \alpha\omega^2$, so the splitting field is $S = \mathbb{Q}(\alpha, \alpha\omega, \alpha\omega^2) = \mathbb{Q}(\alpha, \omega)$.

The polynomial $x^3 - 2$ is irreducible over $\mathbb{Q}$ (Eisenstein at 2), so $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$. Next, $\omega \notin \mathbb{Q}(\alpha) \subseteq \mathbb{R}$ and ω is a root of x^2+x+1, which is therefore its minimal polynomial over $\mathbb{Q}(\alpha)$; so $[\mathbb{Q}(\alpha, \omega) : \mathbb{Q}(\alpha)] = 2$. Therefore

$$[S : \mathbb{Q}] = 2 \cdot 3 = \mathbf{6}.$$

Problem 5

Let $L = \mathbb{Q}(2^{1/3}, 3^{1/4})$. Compute $[L : \mathbb{Q}]$.

Solution. The polynomial $x^3 - 2$ is irreducible over $\mathbb{Q}$ (Eisenstein at 2), so $[\mathbb{Q}(2^{1/3}) : \mathbb{Q}] = 3$. The polynomial $x^4 - 3$ is irreducible over $\mathbb{Q}$ (Eisenstein at 3), so $[\mathbb{Q}(3^{1/4}) : \mathbb{Q}] = 4$.

Both subfields lie in L, so by the tower law $3 \mid [L : \mathbb{Q}]$ and $4 \mid [L : \mathbb{Q}]$; since $\gcd(3,4) = 1$, $12 \mid [L : \mathbb{Q}]$. On the other hand, $L = \mathbb{Q}(2^{1/3})(3^{1/4})$ and $3^{1/4}$ is a root of $x^4 - 3 \in \mathbb{Q}(2^{1/3})[x]$, so

$$[L : \mathbb{Q}] = [L : \mathbb{Q}(2^{1/3})]\,[\mathbb{Q}(2^{1/3}) : \mathbb{Q}] \leq 4 \cdot 3 = 12.$$

Hence $[L : \mathbb{Q}] = \mathbf{12}$. ■

Problem 6

Which are normal extensions of $\mathbb{Q}$? (Recall that a finite extension is normal iff it is the splitting field of some polynomial over the base field; equivalently, every irreducible polynomial over $\mathbb{Q}$ with a root in the field splits in it.)

(a) $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ **is normal**: it is the splitting field of $(x^2-2)(x^2-3)$ over $\mathbb{Q}$.

(b) $\mathbb{Q}(2^{1/4})$ **is not normal**: $x^4 - 2$ is irreducible over $\mathbb{Q}$ (Eisenstein at 2) and has the root $2^{1/4}$ in $\mathbb{Q}(2^{1/4}) \subseteq \mathbb{R}$, but it also has the non-real root $i\,2^{1/4}$, so it does not split in $\mathbb{Q}(2^{1/4})$.

(c) $\mathbb{Q}(\alpha)$ with $\alpha^4 - 10\alpha^2 + 1 = 0$ **is normal**. Note that for $\beta = \sqrt{2} + \sqrt{3}$ we have $\beta^2 = 5 + 2\sqrt{6}$, so $(\beta^2 - 5)^2 = 24$, i.e. $\beta^4 - 10\beta^2 + 1 = 0$. By symmetry under sign changes, the four roots of $x^4 - 10x^2 + 1$ are $\pm\sqrt{2} \pm \sqrt{3}$, and these are all roots (the degree is 4). So any root has the form $\alpha = \varepsilon_1\sqrt{2} + \varepsilon_2\sqrt{3}$ with $\varepsilon_i \in \{\pm 1\}$.

We claim $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Clearly $\mathbb{Q}(\alpha) \subseteq \mathbb{Q}(\sqrt{2}, \sqrt{3})$. Compute

$$\alpha^3 = 11\varepsilon_1\sqrt{2} + 9\varepsilon_2\sqrt{3}, \qquad \text{so} \qquad \alpha^3 - 9\alpha = 2\varepsilon_1\sqrt{2}.$$

Hence $\sqrt{2} = \varepsilon_1(\alpha^3 - 9\alpha)/2 \in \mathbb{Q}(\alpha)$, and then $\varepsilon_2\sqrt{3} = \alpha - \varepsilon_1\sqrt{2} \in \mathbb{Q}(\alpha)$, so $\sqrt{3} \in \mathbb{Q}(\alpha)$. Thus $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$, which is normal by part (a). (Equivalently, $\mathbb{Q}(\alpha)$ is the splitting field of $x^4 - 10x^2 + 1$.)

Problem 7

Galois groups and the corresponding subfields.

(a) x^2+1 over $\mathbb{R}$. The roots are $\pm i$, so the splitting field is $\mathbb{C}$, with $[\mathbb{C}:\mathbb{R}]=2$. Hence $\mathrm{Gal}(\mathbb{C}/\mathbb{R})=\{\mathrm{id},\tau\}\cong\mathbb{Z}/2\mathbb{Z}$, where τ is complex conjugation. The subgroups and their fixed fields are:

Subgroup H	$\mathbb{C}^H$
$\{\mathrm{id}\}$	$\mathbb{C}$
$\mathrm{Gal}(\mathbb{C}/\mathbb{R})$	$\mathbb{R}$

(b) x^3-1 over $\mathbb{Q}$. By Problem 4(a) the splitting field is $\mathbb{Q}(\omega)$ of degree 2 over $\mathbb{Q}$, so $\mathrm{Gal}(\mathbb{Q}(\omega)/\mathbb{Q})=\{\mathrm{id},\tau\}\cong\mathbb{Z}/2\mathbb{Z}$, where $\tau(\omega)=\omega^2=\bar{\omega}$ (the other root of the minimal polynomial x^2+x+1). The subgroups and fixed fields are:

Subgroup H	$\mathbb{Q}(\omega)^H$
$\{\mathrm{id}\}$	$\mathbb{Q}(\omega)$
$\mathrm{Gal}(\mathbb{Q}(\omega)/\mathbb{Q})$	$\mathbb{Q}$

Problem 8

Let G be a group and M a field. A character of G (in M) is a homomorphism $\chi: G\to M^*$.

(a) **Claim.** If $\chi_1,\dots,\chi_k: G\to M^*$ are distinct characters and $a_1,\dots,a_k\in M$ satisfy

$$a_1\chi_1(g)+\cdots+a_k\chi_k(g)=0 \quad \text{for all } g\in G,$$

then $a_1=\cdots=a_k=0$.

Proof. Induction on k.

Case $k=1$. $a_1\chi_1(g)=0$ and $\chi_1(g)\neq 0$ (it lies in M^*), so $a_1=0$.

Inductive step. Let $k\geq 2$ and assume the claim holds for any $k-1$ distinct characters. Suppose $\sum_{i=1}^{k} a_i\chi_i(g)=0$ for all $g\in G$. Since $\chi_1\neq\chi_2$, there is $h\in G$ with $\chi_1(h)\neq\chi_2(h)$. Replacing g by hg and using multiplicativity,

$$\sum_{i=1}^{k} a_i\chi_i(h)\chi_i(g)=0 \quad \text{for all } g\in G. \tag{1}$$

Multiplying the original relation by $\chi_1(h)$ gives

$$\sum_{i=1}^{k} a_i\chi_1(h)\chi_i(g)=0 \quad \text{for all } g\in G. \tag{2}$$

Subtracting (2) from (1), the $i=1$ terms cancel:

$$\sum_{i=2}^{k} a_i\big(\chi_i(h)-\chi_1(h)\big)\chi_i(g)=0 \quad \text{for all } g\in G.$$

By the induction hypothesis applied to the $k-1$ distinct characters $\chi_2, \dots, \chi_k$, we get $a_i(\chi_i(h) - \chi_1(h)) = 0$ for all $i \geq 2$. For $i = 2$ the factor $\chi_2(h) - \chi_1(h)$ is nonzero, so $a_2 = 0$. The original relation now reads $a_1\chi_1 + a_3\chi_3 + \cdots + a_k\chi_k = 0$, a relation among $k-1$ distinct characters, so by the induction hypothesis again $a_1 = a_3 = \cdots = a_k = 0$. Hence all $a_i = 0$. ■

(b) **Claim.** Let $\mathrm{char}(L) \nmid n$, let $\omega \in L$ be a primitive n-th root of unity, and let $M : L$ be Galois with $\mathrm{Gal}(M/L) = \langle\sigma\rangle$ cyclic of order n. Then there is $\alpha \in M$ with

$$\beta = \alpha + \omega\sigma(\alpha) + \omega^2\sigma^2(\alpha) + \cdots + \omega^{n-1}\sigma^{n-1}(\alpha) \neq 0.$$

Proof. Take $G = M^*$ in (a). For $0 \leq i \leq n-1$ let $\chi_i : M^* \to M^*$ be the restriction of σ^i to M^*. Each χ_i is a group homomorphism (indeed σ^i is a field automorphism, so it maps M^* into M^* multiplicatively), i.e. a character of M^*.

These characters are distinct: σ has order exactly n, so $\sigma^0, \dots, \sigma^{n-1}$ are n distinct automorphisms of M; two automorphisms that agree on M^* also agree at 0, hence would be equal on all of M. So $\chi_0, \dots, \chi_{n-1}$ are n distinct characters.

Now take coefficients $a_i = \omega^i \in M$. They are not all zero (for instance $a_0 = 1$; in fact all are nonzero since $\omega \neq 0$). If we had

$$\sum_{i=0}^{n-1} \omega^i\sigma^i(\alpha) = \sum_{i=0}^{n-1} a_i\chi_i(\alpha) = 0 \quad \text{for every } \alpha \in M^*,$$

then part (a) would force $a_0 = \cdots = a_{n-1} = 0$, contradicting $a_0 = 1$. Hence there is $\alpha \in M^*$ (so $\alpha \in M$) such that

$$\beta = \sum_{i=0}^{n-1} \omega^i\sigma^i(\alpha) \neq 0.$$ ■